

MD SANAUL HAQUE JOY

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📍 Dhaka, Bangladesh 🌐 [linkedin.com/in/sanaul-haque-joy](https://www.linkedin.com/in/sanaul-haque-joy)

PROFILE

A highly motivated Mechanical Engineering graduate from Chittagong University of Engineering & Technology (CUET) seeking a Lecturer position to contribute to engineering education and research. Interested in renewable energy, waste-to-energy conversion, and thermal engineering, with research experience in comparative analysis of ML models for Lithium-ion battery RUL prediction and physics informed feature engineering and a dual branch GRU for cross chemistry prediction of Lithium-Ion battery Remaining Useful Life.

ACADEMIC CREDENTIALS

Bachelor of Science (B.Sc.) in Mechanical Engineering Chittagong University of Engineering and Technology Overall CGPA: 3.58/4.00 Senior year CGPA: 3.82/4.00	2022-2026 Chittagong Bangladesh
Higher Secondary Certificate (HSC) Notre Dame College, Dhaka GPA: 5.00/5.00	2020 Dhaka Bangladesh
Secondary School Certificate (SSC) Motijheel Ideal School and College GPA: 5.00/5.00	2018 Dhaka Bangladesh

RESEARCH INTEREST

Renewable Energy | Automotive Engineering | AI and Reliability Engineering | Material Characterization

RESEARCH EXPERIENCE

Undergraduate Thesis Work: Comparative analysis of machine learning models for Lithium-ion battery RUL prediction.

Objectives:

- To analyze battery degradation data and develop ML based RUL prediction models.
- To compare model performance and evaluate prediction accuracy using performance metrics such as MAE, RMSE, and R^2 .
- To identify the most accurate prediction approach.

PUBLICATIONS

- Sultana S, Salam B, and Haque Joy M S, “**Co-pyrolysis of Rice Husk and LDPE using Eggshell as a Catalyst**,” 9th International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE 2026), Khulna University of Engineering & Technology, Khulna, Bangladesh, December 2026. (**Full Paper submitted – under review**).
- Haque Joy M S, Das D P, “**Physics Informed Feature Engineering and a Dual Branch GRU for Cross Chemistry Prediction of Lithium-Ion Battery Remaining Useful Life**.” (Manuscript in preparation).

COMPETITIONS

Block Chain Olympiad: Proposed a block chain system on carbon credit market.

Supply Chain Competition: Proposed a supply chain model on agriculture product of Bangladesh.

SOFTWARE PROFICIENCY

CAD and Simulation

SolidWorks || Ansys ||
AutoCAD

Programming Language

MATLAB || C++ ||
Fortran || Python

Basic Software

Microsoft Word || PowerPoint ||
Excel

AWARDS

- Bangladesh Government Scholarship (General) in the Secondary School Certificate exam (SSC).
- Bangladesh Government Scholarship (General) in the Primary School Certificate exam.

Certification

- Certified SolidWorks Associate (CSWA) in Mechanical Design.
- Matlab Onramp

PROFESSIONAL EXPERIENCE

Industrial Attachment Trainee

Coats Bangladesh Ltd.

12/2025 – 1/2026

Chittagong
Bangladesh

- **Activity:** Got familiar with Industrial Engineering (IE) Workflows, production scheduling and job sequencing. Learned about energy consumption and thermal operational processes across Power Generation, Dyeing, Winding, and Drying units.

Industrial Visit

S.Alam Cold Rolled Steel Mills Ltd

June 2024

Chittagong
Bangladesh

- **Activity:** Gained comprehensive knowledge of the Combined Cycle Power Plant (CCPP) operations, including working principles of Gas Turbines, Steam Turbines, and HRSG.

Digital Skills Trainee

Enhancing Digital Government and Economy (EDGE) Project
Bangladesh Computer Council (BCC)

09/2024 – 12/2024

Chittagong
Bangladesh

- **Activity:** Completed a 60 - hour training on Computer Fundamentals and Office Applications, gaining practical knowledge of computer applications, digital tools, and office productivity software.

EXTRA-CURRICULAR ACTIVITIES

General Member

CUET CAD Society

Vice President (LWS)

Notre Dame Science Club

General Member

Supply Chain & Business Alliance, CUET

General Member

CUET Automotive and Mobility Society - CAMS, CUET

REFERENCES

1. **Dr Prasanjit Das**
Professor, Department of ME, CUET
Email: prasanjit@cuet.ac.bd, Contact: 01642-997907

2. Dr Jamal Uddin Ahamed

Professor, Department of ME, CUET

Email: jamal@cuet.ac.bd, Contact: 01720-641636

DECLARATION

I hereby declare that the information provided in this CV is true and accurate to the best of my knowledge and belief.



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